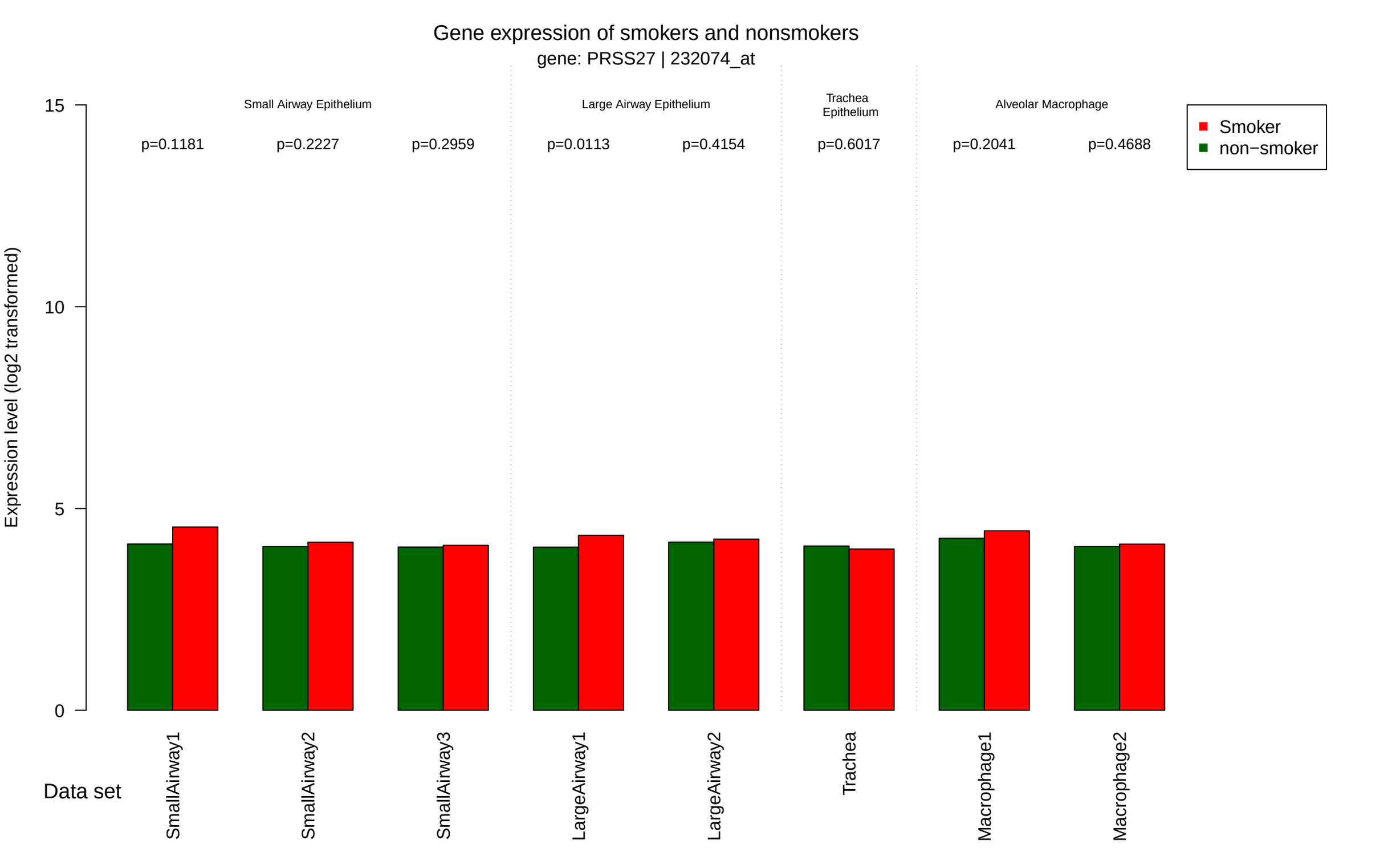
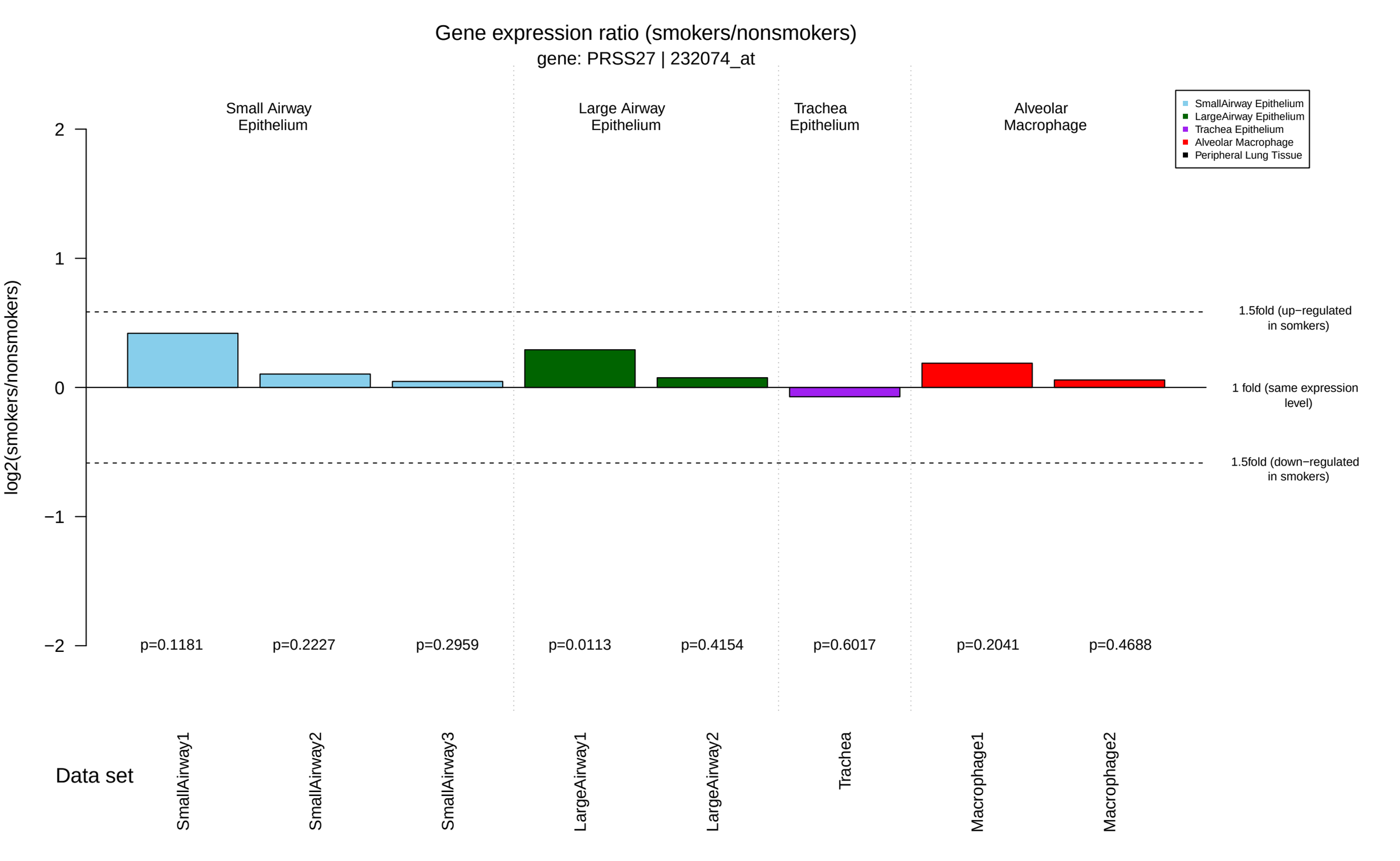
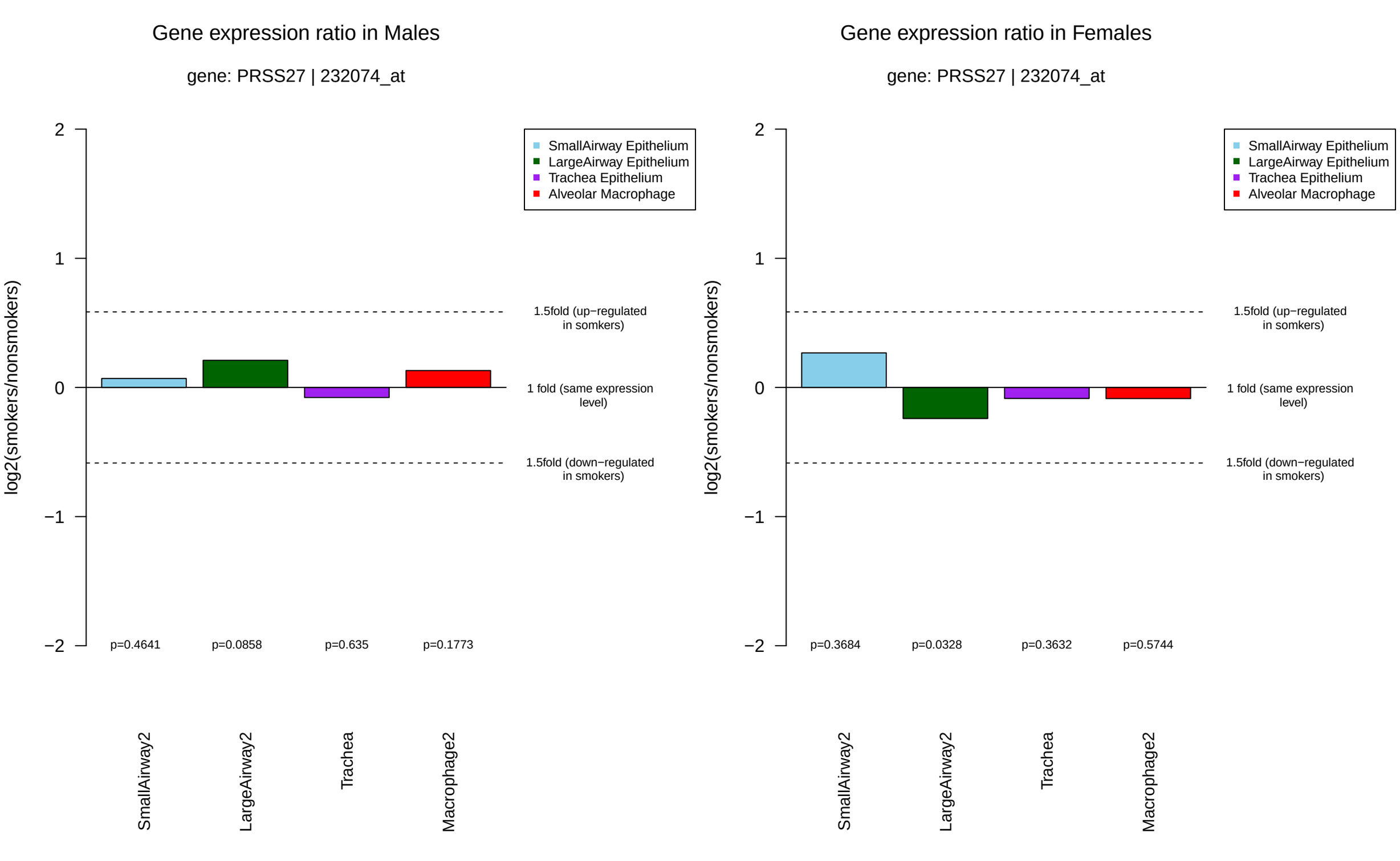
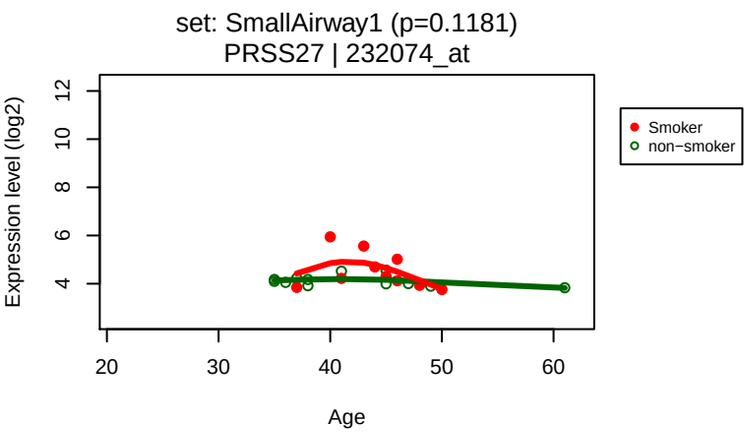
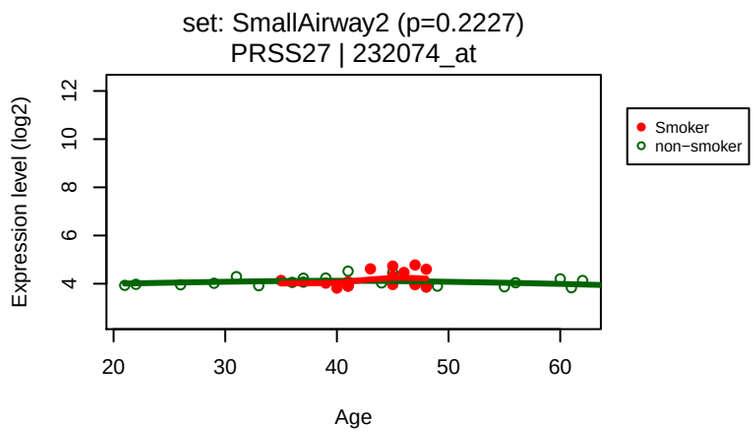
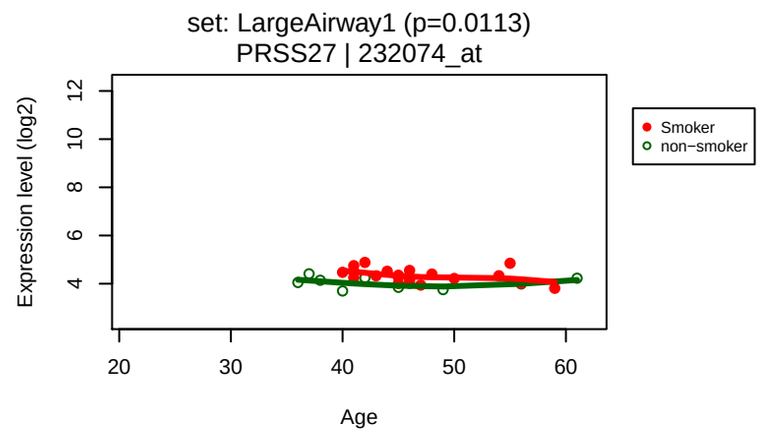

Gene expression of smokers and nonsmokers
gene: PRSS27 | 232074_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1181
p=0.2227
p=0.2959
p=0.0113
p=0.4154
p=0.6017
p=0.2041
p=0.4688
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

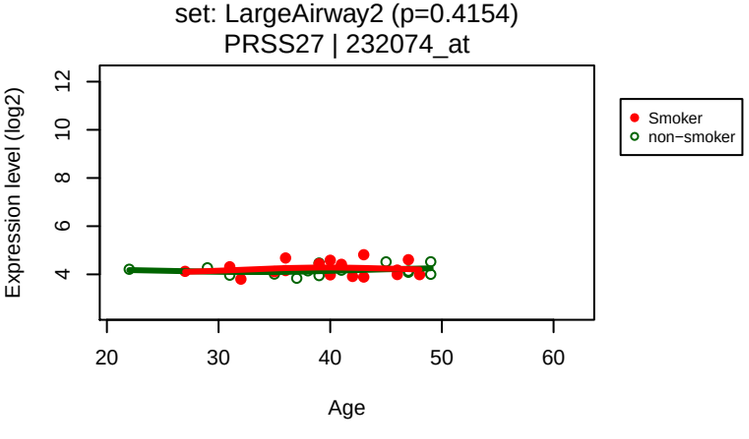

Gene expression ratio (smokers/nonsmokers)
gene: PRSS27 | 232074_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1181
p=0.2227
p=0.2959
p=0.0113
p=0.4154
p=0.6017
p=0.2041
p=0.4688
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

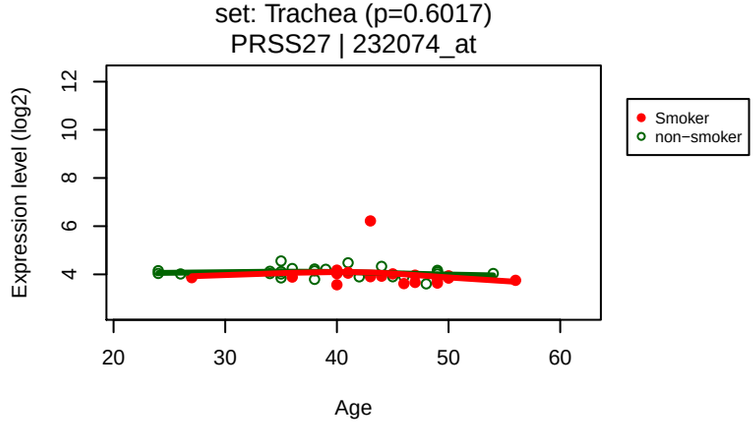

Gene expression ratio in Males
gene: PRSS27 | 232074_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4641
p=0.0858
p=0.635
p=0.1773
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRSS27 | 232074_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3684
p=0.0328
p=0.3632
p=0.5744
SmallAirway2
LargeAirway2
Trachea
Macrophage2

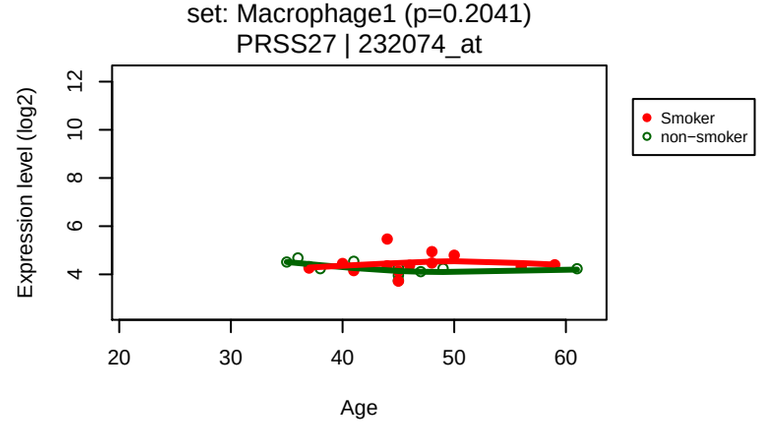

set: SmallAirway1 (p=0.1181)
PRSS27 | 232074_at
Expression level (log2)
Age
Smoker
non−smoker

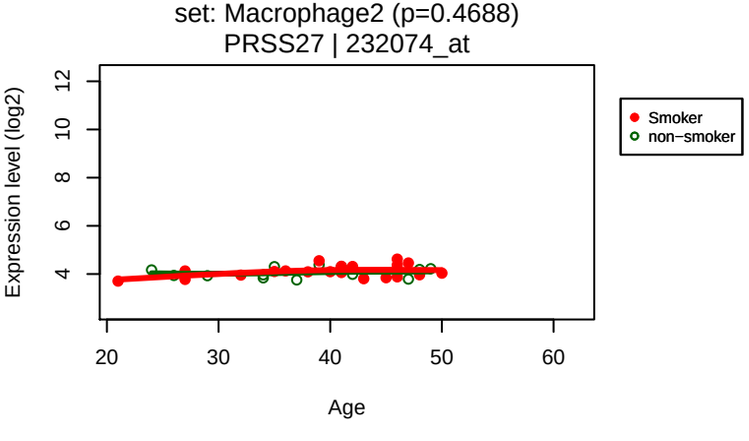

set: SmallAirway2 (p=0.2227)
PRSS27 | 232074_at
Expression level (log2)
Age
Smoker
non−smoker

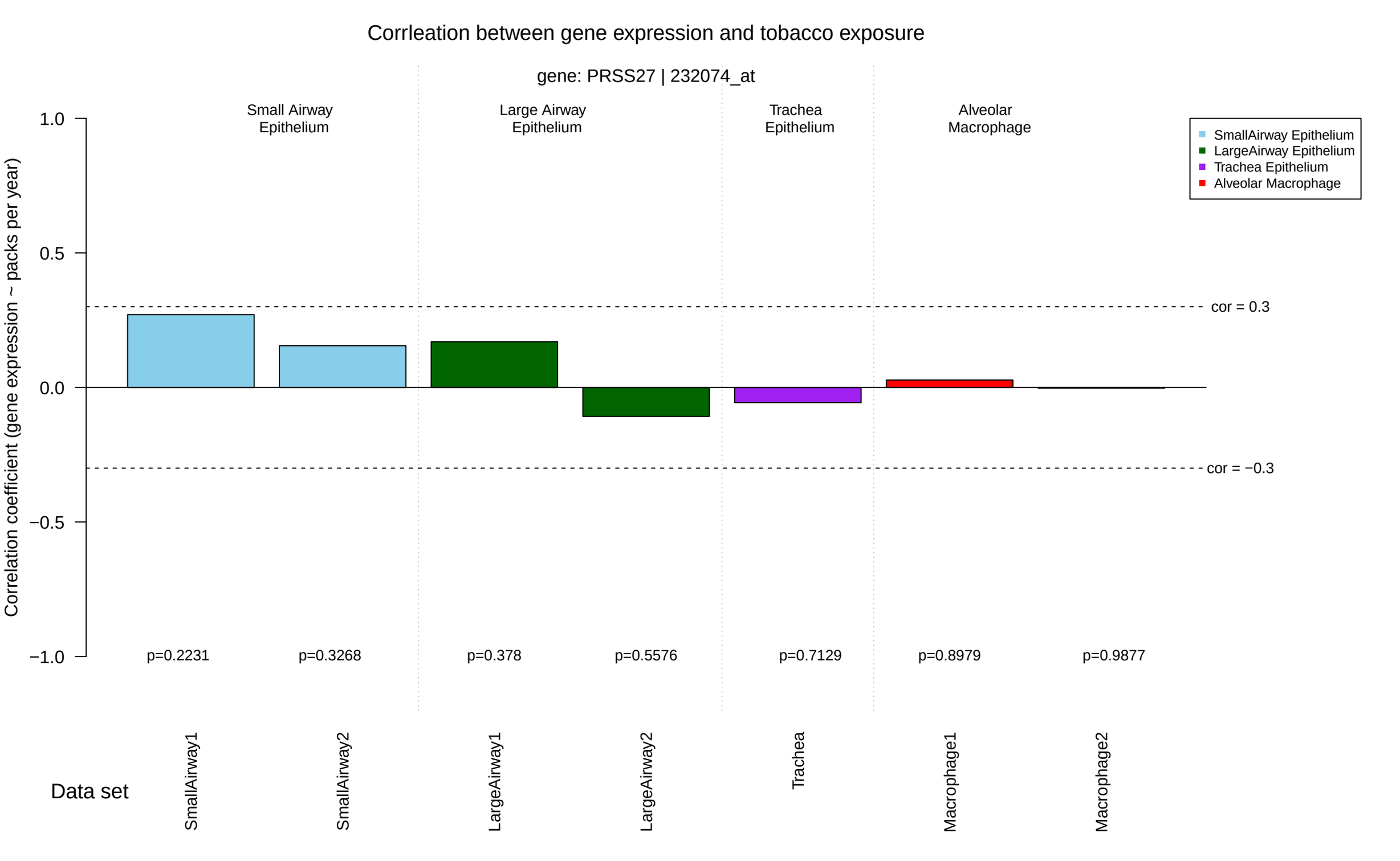

set: LargeAirway1 (p=0.0113)
PRSS27 | 232074_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4154)
PRSS27 | 232074_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6017)
PRSS27 | 232074_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2041)
PRSS27 | 232074_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4688)
PRSS27 | 232074_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRSS27 | 232074_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2231
p=0.3268
p=0.378
p=0.5576
p=0.7129
p=0.8979
p=0.9877
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2